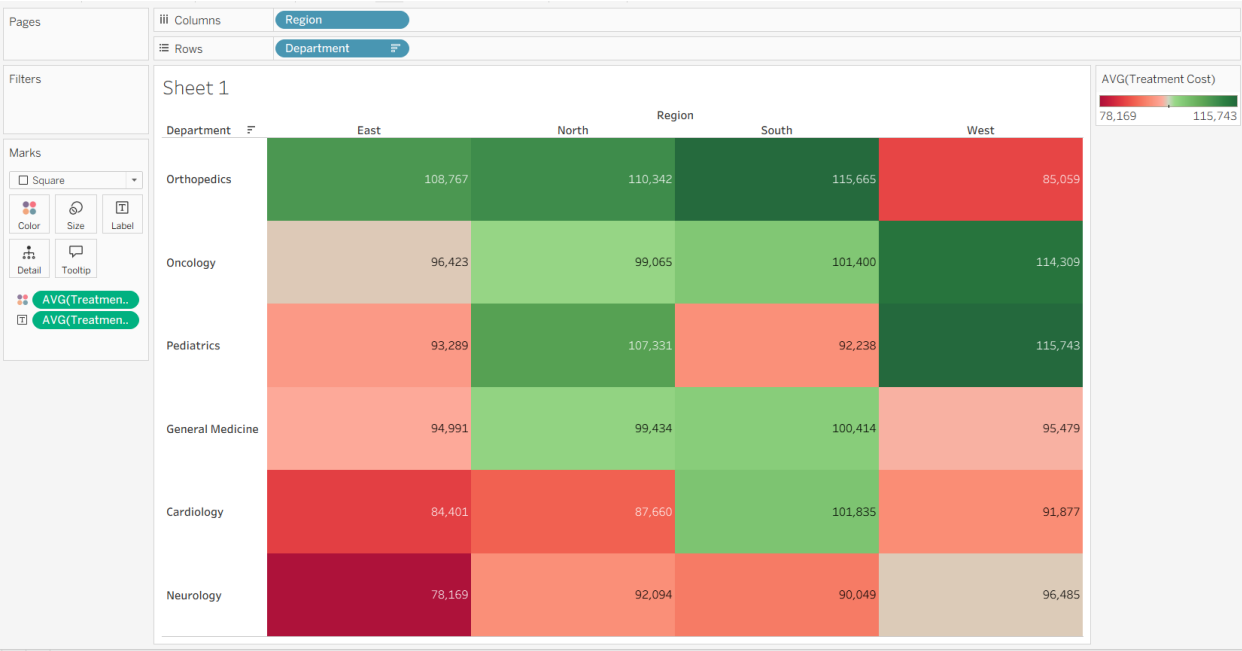


Advanced Visuals in Tableau

Question 1: Create a Heatmap to analyze Treatment Cost by Department and Region.

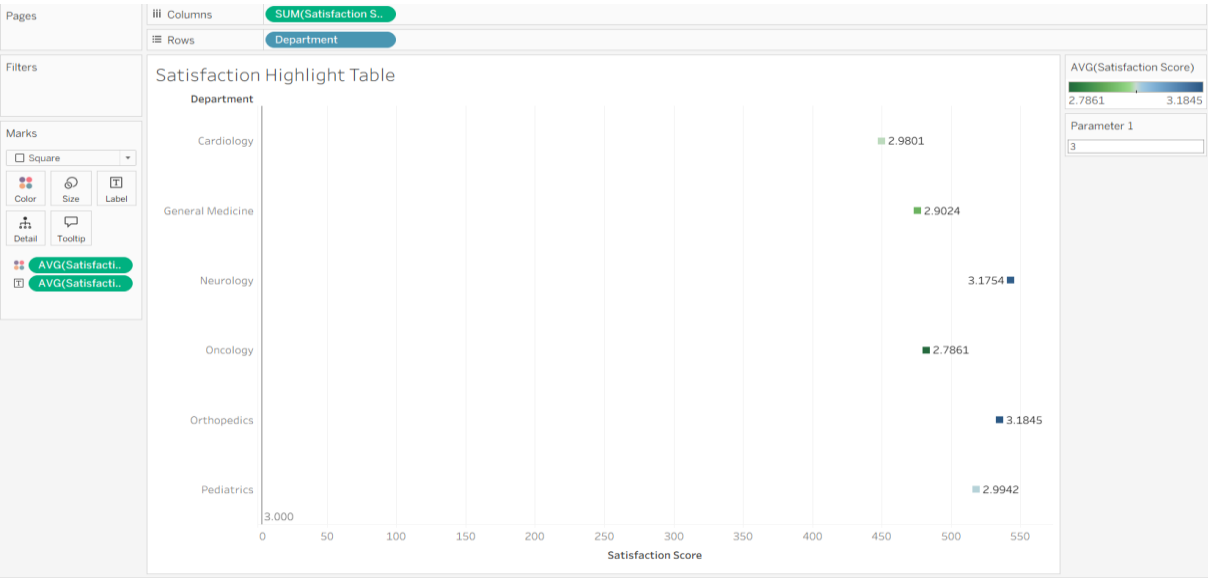
Answer:



The Heatmap instantly reveals that **Pediatrics West (\$115,743)** and **Orthopedics South (\$115,665)** are the most expensive combinations demanding immediate cost review — while **Neurology East (\$78,168)** is the most cost-efficient combination that should be used as a benchmark across the entire healthcare system.

Question 2: Create a Highlight Table showing Satisfaction Score by Department.

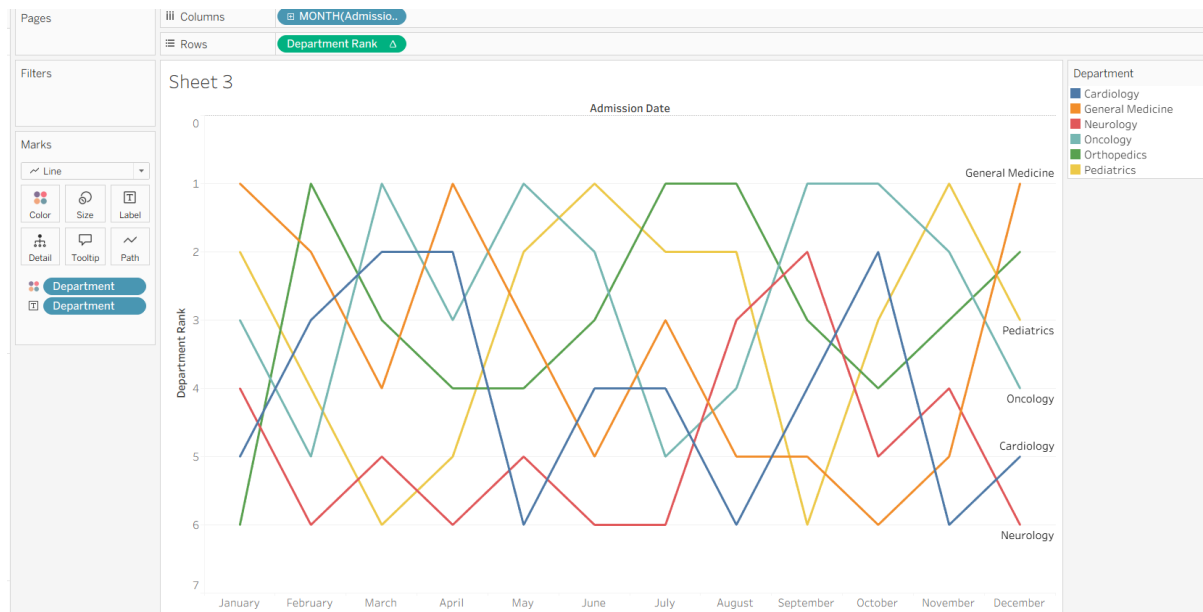
Answer:



The Highlight Table instantly reveals that **Neurology and Orthopedics lead satisfaction at 3.18** while **Oncology (2.79) and General Medicine South (2.48)** are critical problem areas giving hospital management a clear color-coded roadmap of exactly **where patient experience needs urgent improvement**.

Question 3: Create a Bump Chart to show ranking of Departments based on Treatment Cost over months.

Answer:



The Bump Chart reveals that **Oncology dominates the top cost ranking** while **Neurology consistently stays at the bottom** — and most importantly **General Medicine's extreme volatility** (swinging between rank 1 and rank 6) signals an urgent need for cost standardization across that department.

Question 4: Create a Butterfly Chart comparing Male vs Female patient count by Department.

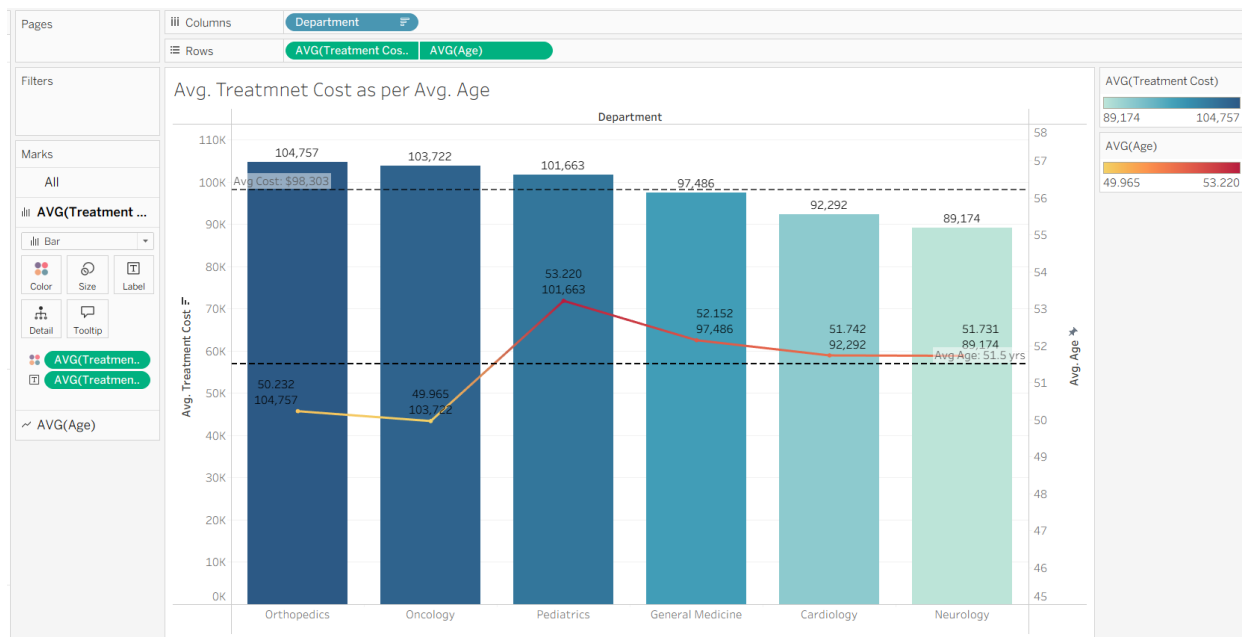
Answer:



The Butterfly Chart reveals **Cardiology is the only female-dominant department** while **General Medicine has the largest gender gap with 20 more male patients** — overall the hospital shows a relatively balanced gender distribution with males slightly outnumbering females across all departments.

Question 5: Create a Dual Axis Chart to compare Average Age and Treatment Cost by Department.

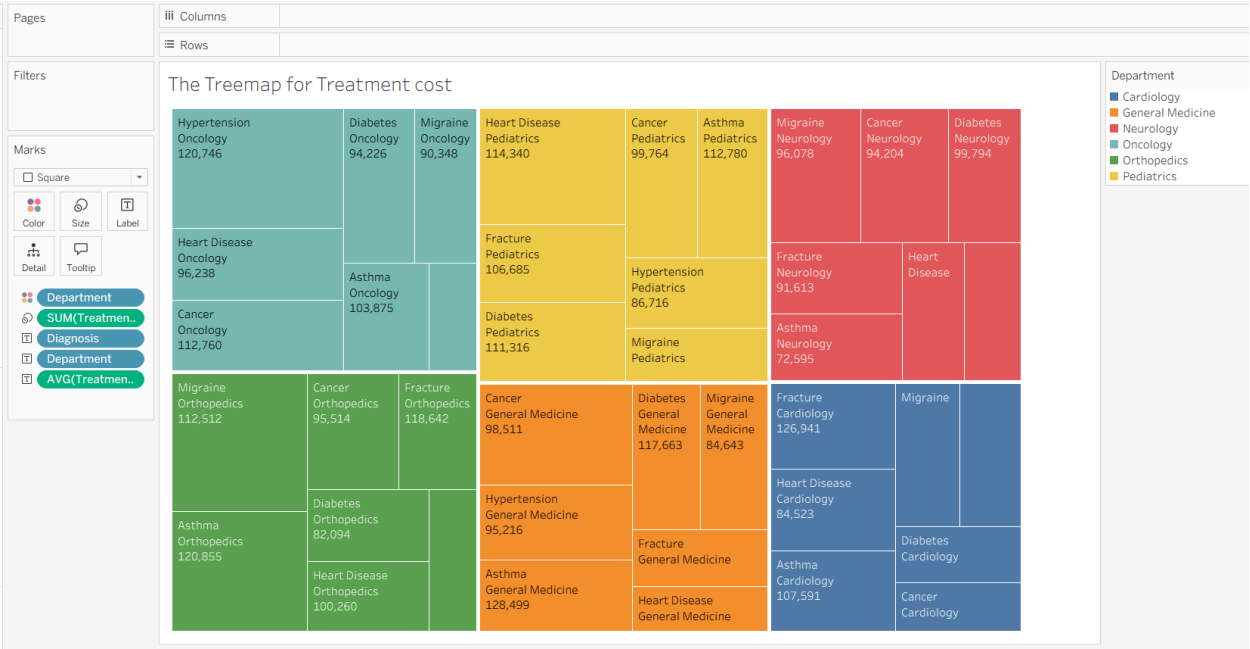
Answer:



The Dual Axis Chart powerfully reveals that **age does NOT drive treatment costs** (correlation = -0.381) — instead **Orthopedics and Oncology are expensive due to treatment complexity** while **Neurology delivers the most cost-efficient care** regardless of patient age — giving management clear direction to focus cost reduction on procedures rather than patient demographics.

Question 6: Create a Sunburst Chart showing Department → Diagnosis hierarchy by Treatment Cost.

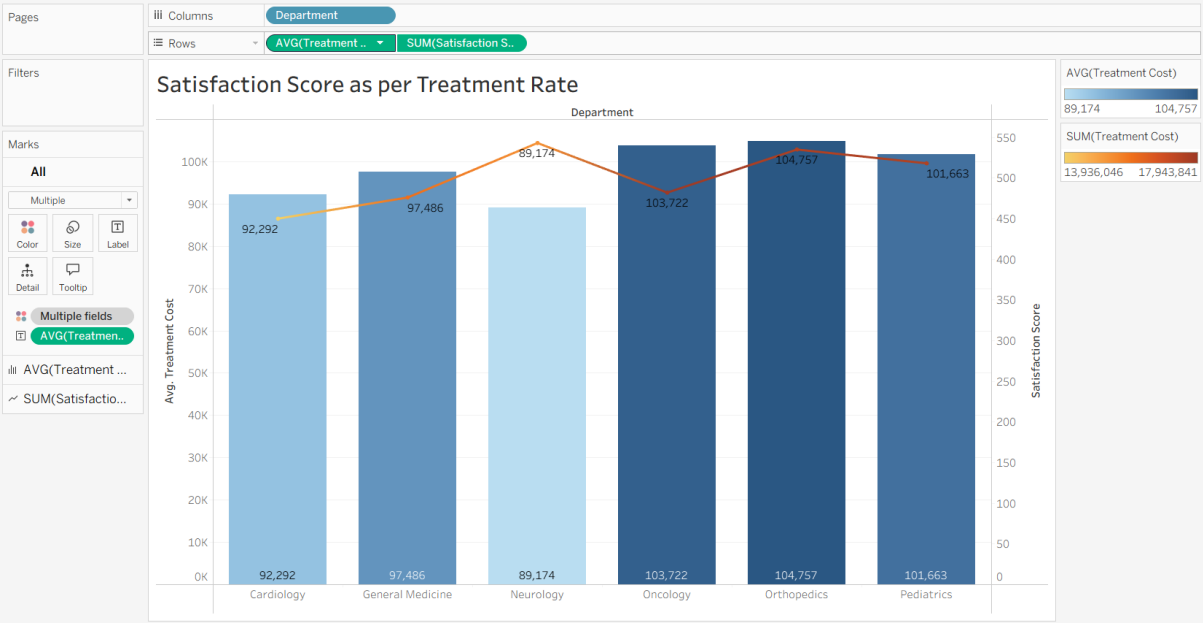
Answer:



The Treemap hierarchy reveals the shocking finding that **the same Asthma diagnosis costs \$128,499 in General Medicine but only \$72,595 in Neurology** a \$56,000 gap for the same disease proving that **treatment protocol standardization** could save millions across the healthcare system.

Question 7: Create a Dual Axis Chart comparing Treatment Cost and Satisfaction Score by Department.

Answer:



The Dual Axis Chart powerfully exposes that **spending more does NOT mean happier patients** (correlation -0.295) **Neurology delivers the best patient value** being both the cheapest AND most satisfying department, while **Oncology is the biggest concern** paying the 2nd highest cost yet receiving the lowest satisfaction score in the entire hospital.

Question 8: Create a Dashboard combining:

- Heatmap
- Dual Axis Chart
- Sunburst Chart
- Butterfly Chart

Answer:

